Wildflower margins and wild-bee pollination in oilseed rape

Aino Laine[1], Joseph Mbeki[2], Clara Ruiz[1]
[1] Department of Ecology, Northfield University [2] Institute for Agroecology, Halden

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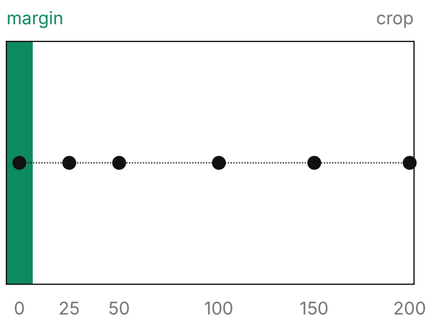
01 INTRODUCTION

Wild bees pollinate most insect-pollinated crops, often better than honey bees [1, 2]. Large uniform fields leave them little to eat outside the short crop bloom [3].

Flower margins are subsidised across Europe, but whether their bees reach the middle of the field is unclear [4].

02 METHODS

36 oilseed rape fields in southern Finland, matched in pairs by size and soil. One field of each pair had a 6 m sown wildflower margin along one edge.



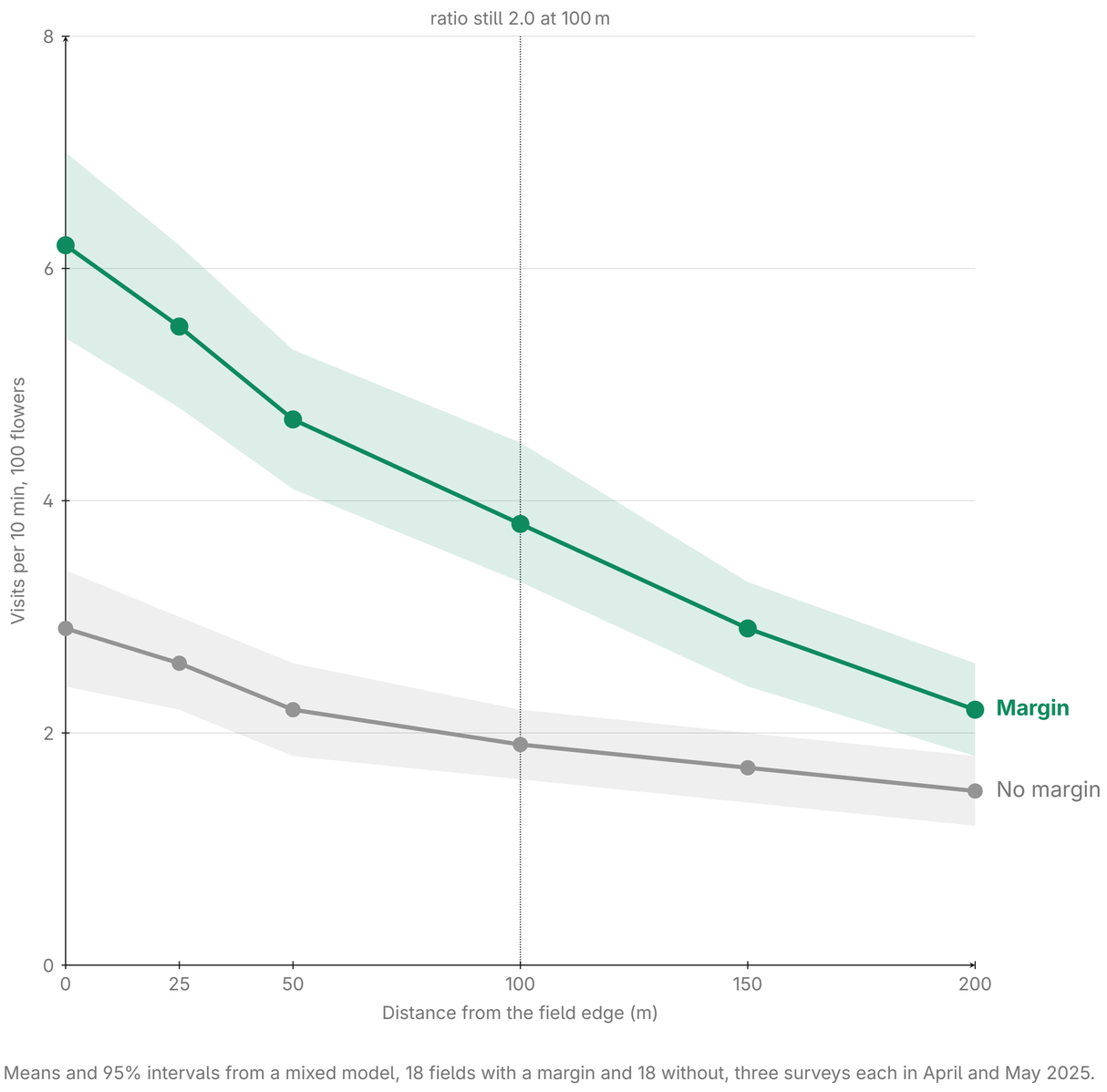
Stations along a transect, metres from the edge.

At each station we counted wild-bee visits to 100 flowers for 10 minutes, three times during bloom. Bees were identified to genus on the wing.

MAIN FINDING

Fields with a wildflower margin get twice as many wild-bee visits, 100 metres deep into the crop.

WILD-BEE VISITS ALONG THE TRANSECT



Means and 95% intervals from a mixed model, 18 fields with a margin and 18 without, three surveys each in April and May 2025.

03 RESULTS IN DETAIL

Distance	Margin	None	Ratio
0 m	6.2	2.9	2.1
25 m	5.5	2.6	2.1
50 m	4.7	2.2	2.1
100 m	3.8	1.9	2.0
150 m	2.9	1.7	1.7
200 m	2.2	1.5	1.5

Visits per 10 min per 100 flowers.

Bumblebees made up 61% of visits, mining bees 27%. Seed set was 9% higher within 50 m of a margin ($p = 0.02$).

04 LIMITS

One crop, one region and one season. Margins were three years old; younger margins may hold fewer bees.

05 REFERENCES

[1] A.-M. Klein et al. *Proceedings of the Royal Society B* 274 (2007), pp. 303–313.
[2] L. A. Garibaldi et al. *Science* 339 (2013), pp. 1608–1611.
[3] C. Kremen et al. *Proceedings of the National Academy of Sciences* 99 (2002), pp. 16812–16816.
[4] M. Albrecht et al. *Ecology Letters* 23 (2020), pp. 1488–1498.

06 CONTACT

a.laine@example.org

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Scan for the full paper, data and code.

https://example.org/wildflower-margins